

# Condition-controlled loops

Revise 360 lesson plan — 60 minutes

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Unit            | Beyond OCR J277: a Python pathway supporting 2.1, 2.2 and 2.3                                                     |
| Lesson          | Condition-controlled loops (lesson 5 of the unit)                                                                 |
| Lesson ID       | pr-105                                                                                                            |
| Experience ID   | pr-105                                                                                                            |
| Plan ID         | PLAN-PR-L05                                                                                                       |
| Content version | 7809b925b693                                                                                                      |
| Website         | <a href="https://revise360.co.uk/experience.html?id=pr-105">https://revise360.co.uk/experience.html?id=pr-105</a> |

## Big question, outcomes and success criteria

**Big question** — How do you decide between a for loop and a while loop before you write a single line of code?

### Prerequisites

The lesson before this one is pr-104. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: pr-101, pr-102, pr-103.

### Assessable outcomes

- pr-105-o1** — know how to write a while loop and what makes it stop repeating
- pr-105-o2** — understand how a validation loop keeps asking until the value typed in is acceptable
- pr-105-o3** — know how a sentinel value ends a loop when the amount of data is not known in advance

### Success criteria the pupil can check against

Taken from the briefs the independent activities themselves show, so these are the words the pupil sees. Their work has met the outcome when it does these things.

- Change only the number in the condition. **pr-105-s1-a3**
- Display the numbers 1, 2, 3, 4, 5 and 6, one on each line. **pr-105-s1-a3**
- The user types in one whole number, which is never less than **1**. **pr-105-s1-a4**
- Replace \_\_\_\_ with one instruction of your own. **pr-105-s1-a4**
- Display the numbers from the one typed down to **1**, one on each line. **pr-105-s1-a4**
- The number typed in is displayed as well, on the last line. **pr-105-s1-a5**
- Use a while loop, not a for loop. **pr-105-s1-a5**
- The user types in one whole number, which is never less than **2**. **pr-105-s1-a6**

## Vocabulary, resources and preparation

**Key vocabulary** — Condition-controlled iteration, Validation loop, Sentinel value, Infinite loop

### Resources

|                   |                                                           |
|-------------------|-----------------------------------------------------------|
| SLIDES            | presentations/PR_L05_ConditionControlledLoops_Lesson.pptx |
| WORKSHEET         | worksheets/PR_L05_ConditionControlledLoops_Worksheet.docx |
| ONLINE EXPERIENCE | experience.html?id=pr-105                                 |

## Before the lesson

1. Sign in on the **teacher dashboard** (`teacher.html`) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the Python runtime and the 360 image are already cached.
4. Check headsets or devices are charged and on the school network.

## How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

**Worth knowing before you promise anything.** In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

## Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

| MIN   | PHASE                                     | TEACHER                                                                                      | PUPILS                                                       | EVIDENCE TO CHECK                                                                            |
|-------|-------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 0-5   | <b>Retrieve</b>                           | Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites. | Answer the starter on the worksheet.                         | The starter box on <code>PR_L05_ConditionControlledLoops_Worksheet.docx</code> is filled in. |
| 5-8   | <b>Review the starter</b>                 | Take answers; correct the one most got wrong before moving on.                               | Correct their own starter.                                   | A correction written, not just a tick.                                                       |
| 8-16  | <b>Explain and model</b>                  | The new idea, then the worked example, typed live rather than shown finished.                | Watch the example being built, and say what each line does.  | They can name the line that does the job, not just copy it.                                  |
| 16-19 | <b>Set up and settle</b>                  | Sign-in, launch, and what to record on paper before anyone types.                            | Sign in, open the lesson, read the task before typing.       | Everyone is at station 1 activity 1, not typing yet.                                         |
| 19-48 | <b>Online: work the stations in order</b> | Pupils work; you circulate and read the editor over shoulders.                               | Work the stations in order, running and checking as they go. | The station badges fill in; the editor shows their own program, not the starter.             |
| 48-53 | <b>Check understanding</b>                | The hinge question, hands down, then the reteach if the class splits.                        | Answer the hinge question, hands down.                       | The split across the options, before any reteach.                                            |

| MIN   | PHASE                                  | TEACHER                                                        | PUPILS                                        | EVIDENCE TO CHECK                           |
|-------|----------------------------------------|----------------------------------------------------------------|-----------------------------------------------|---------------------------------------------|
| 53-58 | <b>Apply or extend</b>                 | The challenge activity, or the exam practice on the worksheet. | The Challenge activity, or the exam practice. | worksheet, Exam practice a) to c)           |
| 58-60 | <b>Exit and set up the next lesson</b> | The exit question and what to finish before next time.         | Write the exit answer.                        | One sentence that answers the big question. |

## The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

### 1. A loop with a condition

- while repeats its block for as long as its condition stays True.
- The condition is tested before every pass, so a while loop can run no times at all.
- Common mistake: never changing the condition inside the loop, which leaves it running forever.

**Challenge on this wall:** What has to change inside a while loop for it ever to stop?

### 2. Asking until it is right

- A validation loop reads a value, then asks again for as long as it breaks the rules.
- Read once above the loop, then read again at the bottom of it.
- Common mistake: leaving the second read out, so the same bad value is tested forever.

**Challenge on this wall:** Why does a validation loop need an `input()` in two different places?

### 3. Sentinel values

- A sentinel is an agreed value meaning 'no more data', such as -1 or the word end.
- The loop keeps going until the sentinel arrives, however many values came before it.
- Common mistake: counting the sentinel itself as one of the real values.

**Challenge on this wall:** Why must a sentinel be a value the real data can never take?

### 4. Counting inside a while

- A counter and a running total are set up before the loop and updated inside it.
- The same pass can count the values and test each one against a rule.
- Common mistake: updating the counter after the loop has already finished.

**Challenge on this wall:** How would you count how many of the numbers read reached 40?

### 5. Choosing for or while

- Use for when the number of repeats is known before the loop starts.
- Use while when it depends on the data, or on what the user chooses to type.
- Common mistake: a while loop with its own counter where a for loop would be shorter and safer.

**Challenge on this wall:** Which kind of loop does a program need when it cannot know how many repeats there will be?

### 6. Putting it together

- Set every total and counter up before the loop, and report them after it.
- One while loop can both check what comes in and gather the figures you need.
- Common mistake: reporting inside the loop, which prints a line for every value read.

**Challenge on this wall:** What should a program print when the sentinel is the very first thing typed?

### The example to model

A while loop repeats for as long as its condition stays true. It has three parts. Set the variable before the loop. Test it in the condition. Change it inside the loop. Miss out that last part and the condition never becomes false, so the loop never stops.

```
n = 3
while n > 0:
    print(n)
    n = n - 1
print("Lift off")
```

It displays **3 / 2 / 1 / Lift off**

**Decisions to model out loud, not just the finished answer:** say why you chose that name, why that line comes before the next one, and what you would try first if it did not work. The example deliberately uses different data from the task, so a pupil cannot copy it and be finished.

## Retrieval and the check for understanding

**Starter** — A program sets count = 5, then a while loop runs for as long as count > 0, and the only line inside the loop prints count. Describe what happens when this program runs, and write down the one line that is missing.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

### Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (pr-105-s4-c2), so the wording a pupil sees matches the wording you say.

**What will this program display?**

**Correct:** 2

### What a pupil might choose instead

- 3
- 0

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

**The course's own response to a wrong answer here:** n goes 1, 3, 5. The loop runs twice, because 5 is not below 5, so the counter reaches 2.

**One response, not four.** one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

### Misconceptions this lesson addresses

- never changing the condition inside the loop, which leaves it running forever. [wall panel](#)
- Which of the three parts of a while loop is missing here? [pr-105-s1-a7, a Fix it activity](#)
- leaving the second read out, so the same bad value is tested forever. [wall panel](#)
- An if checks once and then moves on. What checks again and again? [pr-105-s2-a7, a Fix it activity](#)
- counting the sentinel itself as one of the real values. [wall panel](#)
- Which values reach the line that adds to the total? [pr-105-s3-a7, a Fix it activity](#)
- updating the counter after the loop has already finished. [wall panel](#)
- A counter is set up once. What happens if it is set to [0](#) again on every turn? [pr-105-s4-a7, a Fix it activity](#)

## Guided to independent practice

| STAGE     | ACTIVITIES         | WHAT IT IS                                                                                                                |
|-----------|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| Recognise | <a href="#">13</a> | The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.                   |
| Construct | <a href="#">13</a> | Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer. |

Produce      **23**      The pupil writes it from the brief and nothing else.

Challenge      **6**      The hardest activity on each station. It is not optional and it is not extra: it is the last step of the station.

### Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

### Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

## Delivering the online experience

### Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

### The stations, and what to record at each

| #            | STATION                               | ACTIVITIES | RECORD ON PAPER                                                                                                    |
|--------------|---------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------|
| <b>1</b>     | <b>A loop with a condition</b>        | <b>9</b>   | A loop whose condition never becomes False will freeze the program, and is called an...                            |
| <b>2</b>     | <b>Asking until it is right</b>       | <b>9</b>   | A loop that keeps asking until the value is acceptable has a name of its own, and it is the heart of topic 2.3:... |
| <b>3</b>     | <b>Sentinel values</b>                | <b>9</b>   | Sentinel means guard, because the value stands at the end of the data...                                           |
| <b>4</b>     | <b>Counting inside a while</b>        | <b>9</b>   | A while loop with a counter can do anything a for loop can, but a for loop says it more...                         |
| <b>5</b>     | <b>Choosing for or while</b>          | <b>9</b>   | Both for and while are kinds of iteration, so in the exam you must name the kind you have...                       |
| <b>6</b>     | <b>Putting it together</b>            | <b>9</b>   | A loop that reads until a sentinel copes with no data at all, as long as the totals start at...                    |
| <b>final</b> | <b>Final challenge: for or while?</b> | <b>1</b>   | the answer to the station's challenge                                                                              |

## Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

## Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

**Python: the order is not negotiable.** Each activity teaches what the next one needs, so a pupil works through them in order and completes every one. A shorter lesson, homework or a slow worker does not authorise skipping: the pupil stops where they are and resumes at their earliest unfinished question next time, because progress saves after every answer.

A paper activity, a help request or a tick on this plan does not mark an online question complete, and there is no teacher bypass. If a pupil is stuck after the hint, that is the moment to teach them, not to move them on.

## Assessment, evidence and unfinished work

| ID        | QUESTION                                                                                                                                                                                                                               | MARKS | WHERE                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------|
| pr-105-x1 | A validation loop reads a whole number, then repeats for as long as it is outside 1 to 20. When 99 is typed the program prints out of range over and over. State what has been left out of the loop, and where in the loop it belongs. | 2     | worksheet, Exam practice a); deck, Exam practice slide |
| pr-105-x2 | Write a program that reads whole numbers until -1 is typed, then prints how many numbers were read and their total.                                                                                                                    | 5     | worksheet, Exam practice b); deck, Exam practice slide |
| pr-105-x3 | Write a program that keeps asking for a word until yes or no is typed, printing please answer yes or no for every other word, and then prints the word that was accepted.                                                              | 4     | worksheet, Exam practice c); deck, Exam practice slide |

## Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/pr-105.json`, which is not published with the website.

Every programming activity in this lesson is marked by running the pupil's program against its test cases, so the mark is the program's behaviour and not its appearance. The model solutions are in `answers/codebank/pr-105.json`.

## Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What will this program display?** — 1 2 3 4; 1
- **The user types 3, then 7, then 15. What will this program display?** — again 15; again again again 15
- **The user types 5, then 8, then 0. What will this program display?** — 5 8 0; 5
- **What will this program display?** — 3; 0
- **What will this program display?** — for ran while ran neither ran; for ran neither ran

## Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

## Exit, next lesson and homework

### Exit check

One of these, written, not a show of hands:

- Writing a while loop that stops

- Writing a validation loop that asks again
- Using a sentinel value to end a loop

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

### Retrieval link to next lesson

Next is pr-106, String handling. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

### Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

## Shorter lessons, longer lessons and no internet

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### If this lesson takes more than an hour

It will. There are 55 activities across 6 stations, and a beginner does not finish 55 of them in twenty-nine minutes. Plan it as two or three sessions of an hour: stations 1 and 2 in the first, 3 and 4 in the second, 5, 6 and the final challenge in the third. Each session keeps the same starter-explain-model opening; the online block resumes where the class got to. Honest total: two to three hours.

### A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

A shorter lesson still does not authorise skipping a question. The save-and-resume point is wherever the pupil got to; there is no way to mark a question complete without completing it, and no teacher bypass.

### If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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